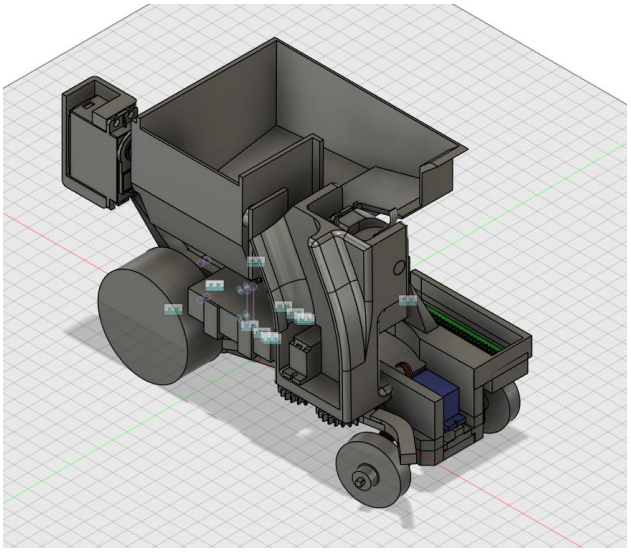


ASME STUDENT DESIGN COMPETITION 2026

OVERVIEW

A remotely controlled robot was built for ASME SDC 2026 to collect and deliver material units in a constrained urban playfield. It uses integrated mechanical, electrical, and Arduino-based control systems to operate within strict size, power, and safety limits.

S. Hibpshman, Z. Jump, & D. Alverado
ASME SDC 2026

ASME Student Design Competition 2026

A remotely controlled robot was built for ASME SDC 2026 to collect and deliver material units in a constrained urban playfield. It uses integrated mechanical, electrical, and Arduino-based control systems to operate within strict size, power, and safety limits.

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April 24, 2026



Table of Contents

List of Figures iv

List of Tables iv

Acronyms Used v

Abstract 1

Foreword 1

Introduction 1

Challenge Statement 2

Design Requirements & Constraints 2

Functional Requirements 2

Physical Constraints 2

Electrical Constraints 2

Operational Constraints 3

Safety Constraints 3

Electrical System Design 3

Power Architecture 3

Custom PCB Integration 4

Power Distribution Board 5

Control System 7

Communication Interface 7

Input Processing 7

Output Control 7

Control Strategy 7

Integration Strategy 8

Chassis Design 8

Drivetrain & Locomotion 12

Rotary Dumping Mechanism 15

Design Tradeoffs Discussion 17

Electrical Systems 17

Competition Performance Results 18

Conclusion 18

Future Improvements 19
Appendix 20
References 22

List of Figures

Figure 1. PCB Board KiCAD Rendering (Hibpshman, 2026) 5
Figure 2. Effect of Internal Resistance on Battery Voltage Under Load (Hibpshman, ANALYSIS OF BATTERY VOLTAGE UNDER LOAD, 2026) 6
Figure 3. Initial Design for the Lower Chassis (Jump, 2026). 9
Figure 4. Full Chassis Assembly: Initial Design (Jump, 2026). 9
Figure 5. Chassis Overhaul: V3 (Jump, 2026). 10
Figure 6. Final Chassis Design Gallery (Jump, 2026; Alvarado, 2026). 12
Figure 7. Steering Assembly: Front View (Jump, 2026). 13
Figure 8. Steering Assembly: Top-Down View (Jump, 2026). 13
Figure 9. Steering Radius Closeup (Jump, 2026). 14
Figure 10. Bottom View, Brushless Motor (Jump, 2026) 14
Figure 11. Yellow Motors: Sideview (Jump, 2026) 15
Figure 12. Assembled Dumping Bucket Top View (Jump, 2026; Alvarado, 2026). 16
Figure 13. Dumping Bucket Pivot Showcase (Jump, 2026). 16
Figure 14. Dumping Bucket Disassembled (Jump, 2026). 17
Figure 15. Arduino Nano Every Code Version 10.0.0 (Hibpshman, 2026; SwapnilNimbalkar, 2020). 20

List of Tables

Table 1. Bill of Materials (Hibpshman, 2026). 21

Acronyms Used

American Society of Mechanical Engineers (ASME) 1
Direct Current (DC) 6
Electrical & Computer Engineering (ECE) ii
Lithium Polymer (LiPo) 17
Mechanical Engineering (MENG) ii
Nickel Metal Hydride (NiMH) 3
Polyethylene Terephthalate Glycol (PETG) 8
Polylactic Acid (PLA) 8
Power Distribution Board (PDB) 3
Printed Circuit Board (PCB) 1
Pulse Width Modulation (PWM) 7
Remote Control (RC) 8
Student Design Competition (SDC) 1

Abstract

This project presents the design and implementation of a compact, remotely controlled robotic vehicle developed for the 2025–2026 American Society of Mechanical Engineers (ASME) Student Design Competition (SDC). The robot functions as an urban-scale waste collection and transport system, operating within a constrained mock “downtown Dallas” environment. The system was required to remain within a 50 cm × 50 cm × 50 cm volume, operate using rechargeable dry-cell batteries, and successfully collect, transport, and sort standardized material units under timed competition conditions.

The robot architecture integrates a custom electrical power distribution system, Printed Circuit Board (PCB)-based control electronics, Arduino-based control logic, and a radio-controlled interface. Mechanical subsystems include a steerable chassis designed for tight urban navigation and a payload system for material handling.

Key engineering challenges included maintaining electrical reliability under high current loads, ensuring robust wireless control in a multi-vehicle RF environment, and achieving consistent steering performance within narrow roadway constraints. The final system demonstrates successful integration of modular electrical and mechanical subsystems optimized for rapid repairability, competition safety requirements, and operational reliability under timed conditions.

Foreword

This document is meant to serve as a formal written accounting of the design choices of each team lead. As such, figures are provided to the reader as a visual aid. Documentation of the robotic arm subsystem, led by David Alvarado, was unavailable at the time of writing and is therefore not included. Where the arm interfaces with the electrical, chassis, or dumping systems, those interfaces are described from the perspective of the subsystem being documented.

Seth Hibpshman is the primary writer of this document, compiling sources, figures, tables, and information from other team members. The sections ‘Chassis Design’, ‘Drivetrain & Locomotion’, and ‘Rotary Dumping Mechanism’ were written solely by Zach Jump.

Introduction

The ASME SDC 2026 challenges teams to design a compact, remotely controlled robotic system capable of operating in a simulated urban environment. The objective is to collect, transport, and sort standardized waste units within a structured playfield representing a simplified “downtown Dallas” layout.

The robot must operate under strict constraints including a maximum bounding volume of 50 cm per side, use of rechargeable dry-cell batteries, and compliance with strict safety and

operational rules. Devices are evaluated in both solo timed runs and head-to-head elimination rounds, with scoring based on efficiency, accuracy, and violation penalties.

This project focuses on developing a reliable and modular robotic platform capable of:

- Navigating narrow urban pathways (10–18 cm street widths)
- Collecting and transporting discrete material units
- Operating under continuous timed constraints without pause
- Maintaining control via a single radio transmitter
- Surviving repeated handling, resets, and rapid repair scenarios

The design emphasizes reliability, maintainability, and electrical robustness under competition conditions.

Challenge Statement

The AY25-26 ASME SDC is inspired by the global issue of recycling and waste collection management. The AY25-26 SDC challenge is to design, fabricate, and test a versatile waste collection device that can rapidly and accurately navigate a model city with various terrains, collect and sort garbage and recycling, and transport and dump the waste materials into the correct facility (recycling center or landfill). This device must operate in a safe and efficient manner that obeys traffic and safety laws and avoids property damage or waste spillage.

Design Requirements & Constraints

Functional Requirements

- Remotely controlled vehicle operation via single operator
- Ability to navigate constrained urban-style pathways
- Collection and transport of discrete waste units
- Delivery of materials to designated zones (landfill / recycling equivalent)
- Continuous operation during timed demonstrations (5–10 min runs)

Physical Constraints

- Maximum bounding volume: 50 cm × 50 cm × 50 cm (including stored tools and batteries)
- Must be fully stowable inside rigid sizing box
- Expansion allowed only after deployment and under own power
- All components must remain inside box during competition when not active

Electrical Constraints

- Powered exclusively by rechargeable dry-cell batteries
- Onboard power distribution required for multiple subsystems

- No external tethering during operation
- Electrical system must support motors, control electronics, and communication system simultaneously

Operational Constraints

- Single operator control via remote transmitter
- No pausing of competition timer during operation
- Minor repairs allowed on clock (battery swaps, wire reconnection)
- Major repairs prohibited during runs (no soldering, reprogramming, rebuilds)

Safety Constraints

- Device must not damage playfield or competition infrastructure
- Unsafe behavior results in immediate disqualification
- Must comply with traffic-style navigation rules in shared environment
- Must avoid interference with other competing robots

Electrical System Design

The robot is powered by a Nickel Metal Hydride (NiMH) rechargeable battery pack as the main energy source. A custom power distribution board splits and regulates power between high-current and low-voltage subsystems, including the drivetrain, control electronics, and radio receiver. The design separates power domains to improve reliability, reduce electrical noise, and manage current spikes for stable operation.

Power Architecture

The robot is powered by a 6-cell NiMH AA battery pack with a nominal voltage of 7.2 V, providing the primary energy source for all subsystems. Power is routed from the battery pack into a custom Power Distribution Board (PDB), which separates and conditions power for high-current and low-voltage subsystems.

From the PDB, unregulated battery voltage is supplied directly to the motor driver to support the drivetrain under peak load conditions. A bulk capacitance stage of 4000 µF (four 1000 µF capacitors) is placed on the high-current rail to mitigate voltage dips and absorb transient current spikes during motor acceleration and steering events.

High-torque servos are powered directly from the main battery rail to ensure sufficient current availability during load-intensive actuation. A regulated low-power branch is routed from the PDB into a buck converter mounted on the custom PCB, stepping the voltage down to 5 V for logic and low-power components.

The 5 V regulated rail powers the Arduino microcontroller, radio receiver, and low-torque servos. This separation between high-current and logic domains improves electrical noise isolation and prevents control system resets during drivetrain load fluctuations.

Overall, the architecture prioritizes reliability under dynamic load conditions by combining direct-drive power paths for actuators with regulated low-noise power for control electronics, supported by local bulk capacitance for transient stability.

Custom PCB Integration

A custom printed circuit board was designed in KiCad and fabricated through JLCPCB to consolidate control, signal routing, and power distribution for the robot. The PCB serves as a central interface between the control electronics, power regulation stage, and external actuators, reducing wiring complexity and improving system reliability under competition conditions.

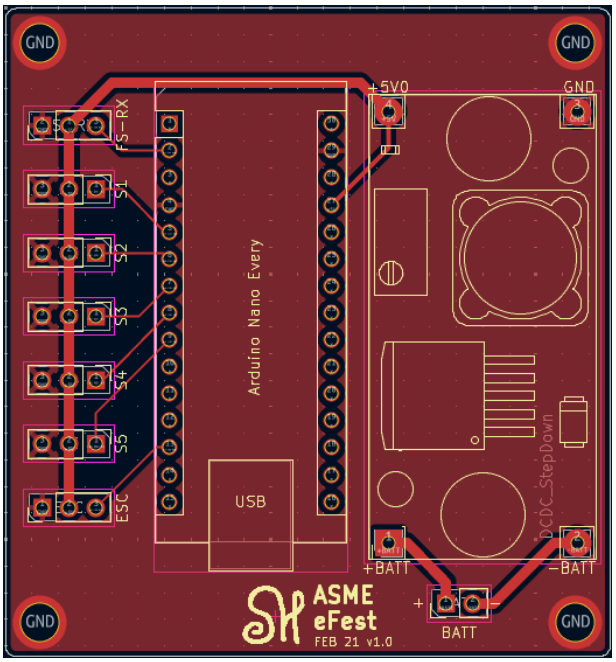
The board integrates a footprint for an LM2596 buck converter module, which provides regulated 5 V power from the main battery rail. This regulated supply powers the Arduino Nano Every, radio receiver, and low-power servos. The Arduino Nano Every is mounted directly onto the PCB to minimize loose connections and improve mechanical robustness during operation.

Signal and power distribution are provided through five male pin header interfaces. These include dedicated breakout headers for control signals as well as separated ground and power rails arranged in a 5×3 configuration to support modular subsystem connections. This layout allows flexible interfacing with external components using either Dupont connectors for modularity or direct soldered connections for high-reliability links in critical subsystems.

A continuous ground plane was implemented across the PCB to improve signal integrity and reduce noise coupling between high-current motor circuits and sensitive logic signals. This design choice helps mitigate interference introduced by switching loads such as motors and servos.

The custom PCB reduces wiring density, improves electrical reliability, and provides a structured interface between power, control, and actuation subsystems while maintaining flexibility for rapid repair and reconfiguration during competition conditions.

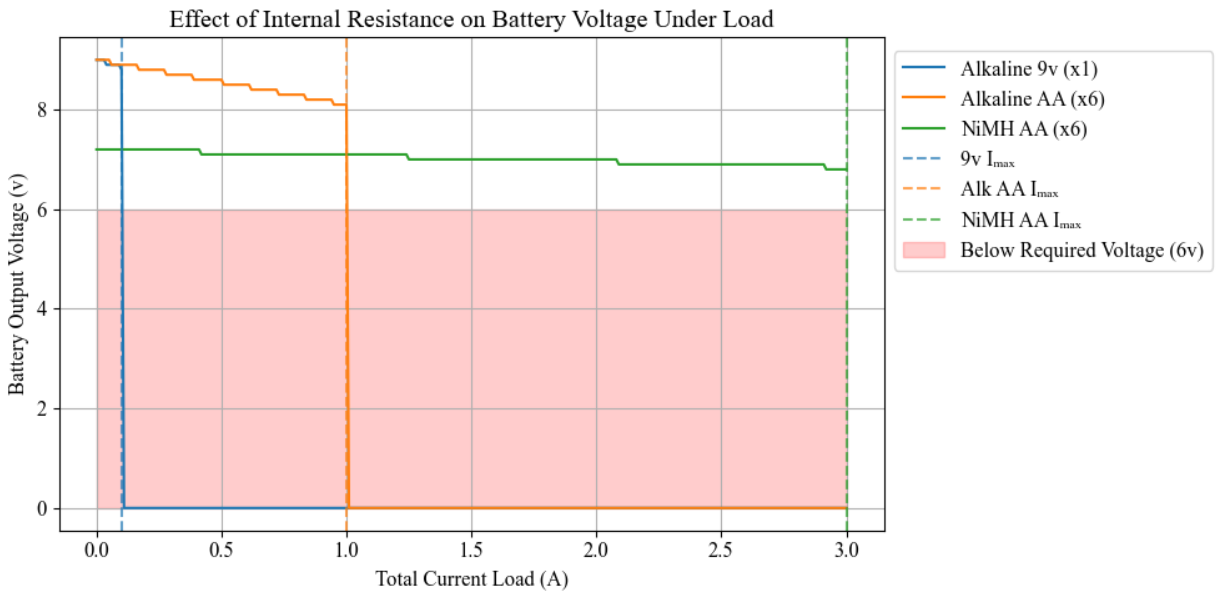
Figure 1. *PCB Board KiCAD Rendering (Hibpshman, 2026).*



Power Distribution Board

A custom power distribution board was designed to manage high-current power delivery and improve electrical reliability across the robot. The PDB serves as the primary interface between the 7.2 V NiMH battery pack and the robot's major power-consuming subsystems, including the drivetrain, high-torque actuators, and regulated control electronics. The 7.2 NiMH battery pack was chosen due to the chemistry allowing the highest amperes to flow to the brushed motor system; researched by Hibpshman Figure 2.

Figure 2. Effect of Internal Resistance on Battery Voltage Under Load (Hibpshman, ANALYSIS OF BATTERY VOLTAGE UNDER LOAD, 2026)



To improve transient load stability, the board incorporates four 1000 µF electrolytic capacitors placed across the main power rail. This bulk capacitance helps buffer voltage drops and absorb current spikes generated during motor startup, direction changes, and steering loads.

The drivetrain is driven through an integrated brushed Direct Current (DC) H-bridge motor driver, supporting bidirectional control (forward and reverse) of the drive motors. This allows precise teleoperated movement within constrained urban pathways while maintaining simplicity and robustness in the power stage.

A physical power switch is included on the PDB to enable quick and reliable system shutdown for safety and handling during setup and repairs.

For actuator distribution, the PDB provides three identical 3-pin header outputs used for high-torque servos. These headers supply raw battery voltage directly from the PDB, while control signals are routed separately from the custom PCB. This separation ensures that high-current servo loads do not interfere with low-power logic systems.

The PDB and control PCB are interconnected via two dedicated soldered wire harnesses, selected over detachable connectors for improved mechanical and electrical reliability under vibration and competition handling conditions. This hardwired connection reduces the risk of intermittent failures during operation.

Overall, the power distribution board centralizes high-current management, improves transient stability through bulk capacitance, and provides a robust separation between power-intensive actuators and sensitive control electronics.

Control System

The robot utilizes an Arduino-based central controller responsible for translating wireless remote inputs into motor and actuator commands. The system is designed for direct teleoperation, prioritizing deterministic response and reliability under competition conditions.

Communication Interface

Wireless control is provided through a FlySky FS-iA6B receiver paired with its corresponding transmitter. The receiver outputs an iBUS serial signal, which is read directly by the Arduino via a hardware serial interface. The iBUS protocol is decoded in software to extract individual channel values representing steering, throttle, and auxiliary control inputs.

Input Processing

Incoming iBUS frames are validated using checksum verification to ensure signal integrity. Once validated, channel data is parsed into normalized pulse-width values (approximately 1000–2000 µs range). These values are then mapped to actuator-specific control ranges.

To improve drivability, the throttle input is filtered using a simple exponential smoothing approach, reducing jitter and preventing abrupt motor commands during rapid stick movements.

Output Control

The Arduino generates control outputs for both brushed motor actuation and multiple servo-based mechanisms:

- **Steering control:** A servo-driven steering actuator receives direct pulse-width commands, limited to a safe operating range for mechanical protection and repeatability.
- **Drive system:** A brushed DC motor is controlled using a bidirectional H-bridge interface. Forward and reverse motion are generated using Pulse Width Modulation (PWM) outputs on separate digital pins, with a deadband region implemented around the neutral input to prevent unintended motion.
- **Auxiliary servos:** Additional channels control subsystem actuators including a gripper mechanism and a shoulder joint. These are mapped from input channels to angular servo positions.

Control Strategy

The control strategy is fully manual (non-autonomous), with real-time operator control of all motion and manipulation functions. A deadband around neutral throttle reduces drift and improves stability during low-speed navigation in constrained environments. Signal smoothing is applied to throttle input to improve mechanical longevity and reduce current spikes in the drivetrain.

Integration Strategy

The control system is tightly integrated with both the power distribution board and custom PCB. The Arduino acts as the central hub, receiving decoded wireless inputs and distributing PWM outputs to motor drivers and servo lines.

High-current actuation (drive motor and high-torque servos) is electrically separated from logic-level control to reduce noise coupling and prevent system resets. Signal integrity is maintained through direct wiring between the receiver, Arduino, and PCB, with critical connections soldered where mechanical reliability is required under vibration and handling stress.

Failsafe behavior is inherently supported through loss-of-signal behavior in the radio receiver, causing output values to return to neutral states, effectively stopping motor output when communication is interrupted.

Chassis Design

The chassis was very challenging to design. The design process was very iterative with multiple design overhauls. At first the chassis was designed like a Remote Control (RC) car with the rest of the systems integrated into the RC car style layout. The idea was that the drivetrain system could be the main system and other systems such as the dumper and the arm could be mounted separately.

The idea was also that the entire chassis would be 3d printed to allow for ultimate design flexibility. Polylactic Acid (PLA) was used at first due to its low cost and because of how easy this material is to print. After a few prototypes the material was scrapped because it was found to be too brittle and would snap before flexing. As a result the material was switched to Polyethylene Terephthalate Glycol (PETG). PETG offers similar strength but is much more ductile, this allowed for damage to only deform the chassis instead of fracturing.

Figure 3. Initial Design for the Lower Chassis (Jump, 2026).

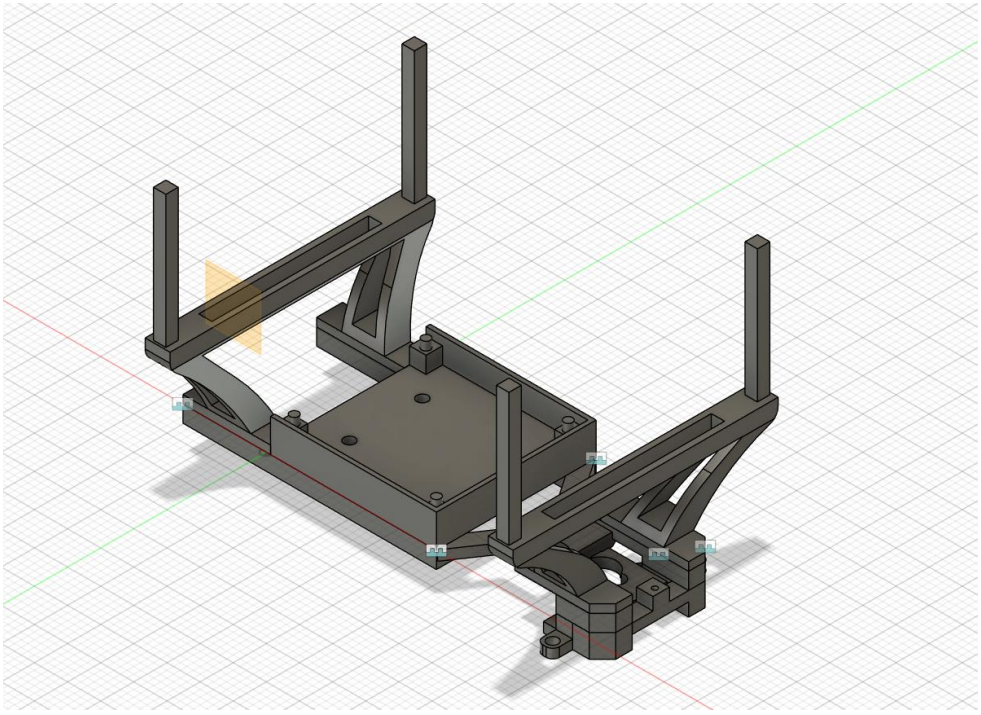
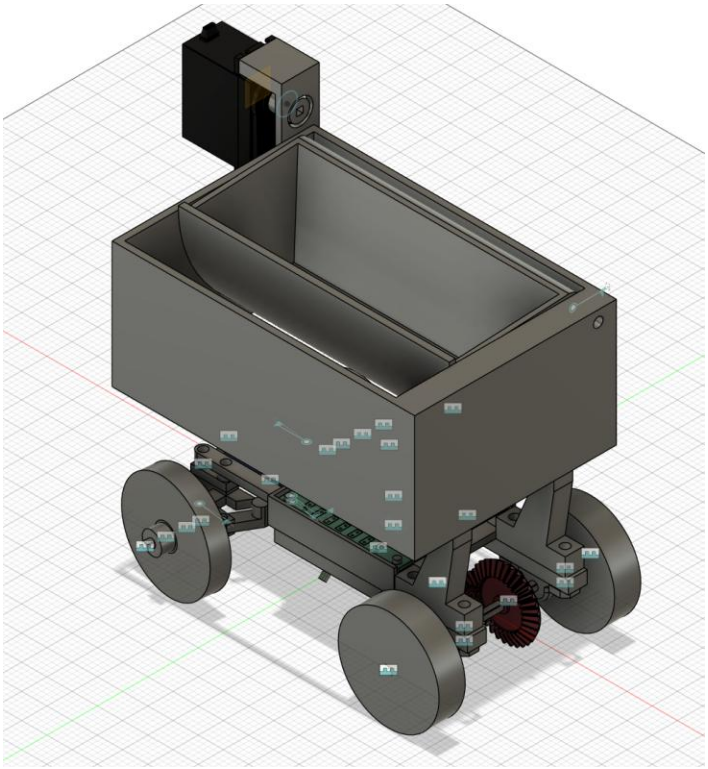


Figure 4. Full Chassis Assembly: Initial Design (Jump, 2026).



At this point, the idea was that the dumping container would take up as much space as possible while allowing proper space for the robot arm to be added into the design. As the arm was designed this design changed.

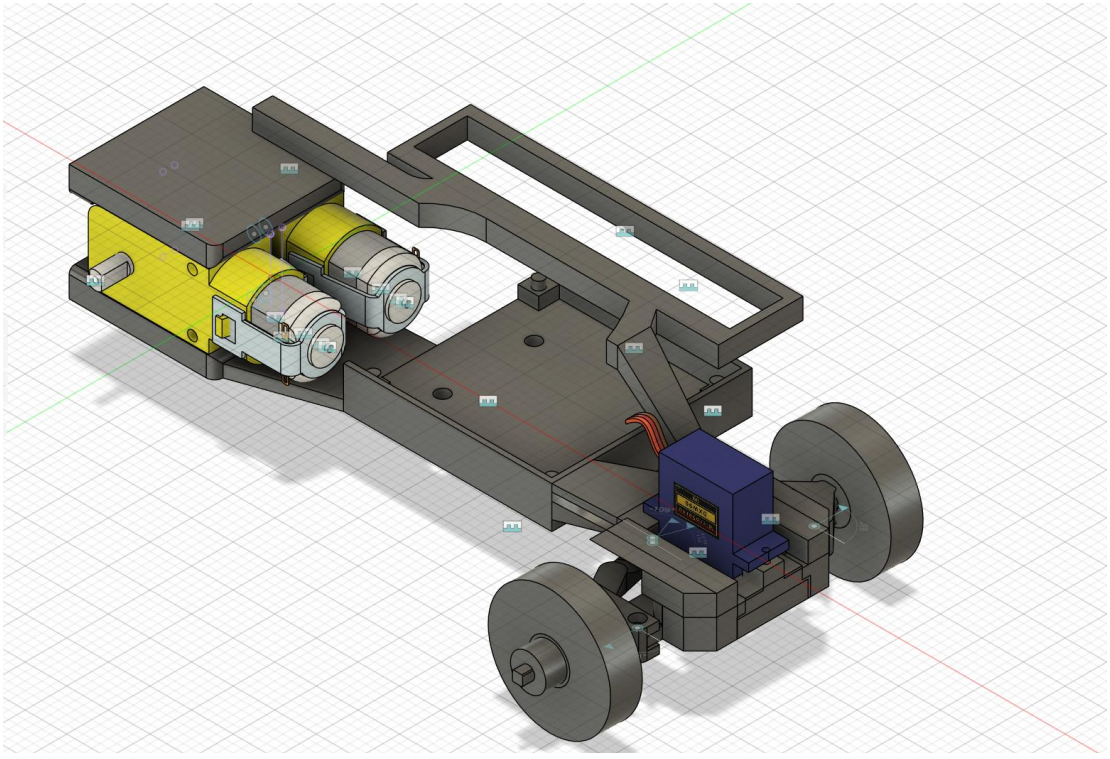
This design was built entirely around what was known about the design in the early development stages. The front half of the chassis features the mounting for the steering of the robot. While the back half was undeveloped due to the rear wheel drive design choice was incomplete at this point.

The top of the chassis provides potential servo mounting points for the dumping mechanism and robotic arm. As well as battery pack mounting. The pins on the bottom half of the chassis would slide into the slots on the top half creating a single assembly. This was the initial chassis design.

After more mechanical and electrical components were designed the chassis design began to change. The overall design went from building parts into the chassis to building the chassis around the parts. At this point, much of the focus was on designing the subsystems such as the robotic arm, electronics, and drivetrain. The chassis remained untouched for a few weeks. Once strong drafts of these systems were designed, it was time to rethink the chassis.

At first, it was challenging to restart the design and focus on both allowing proper space for component installation and maintaining design constraints. Some notable changes that took place in those weeks away from chassis design was a change in the drive system to DC yellow motors and that the battery pack sizing was developed.

Figure 5. Chassis Overhaul: V3 (Jump, 2026).



The chassis had begin to change from two separate halves to a single system that fit subsystems into a constrained space. The Arduino mounting had shifted much lower to the ground and towards the outside edge to allow space for the robotic arm to mount into place. The chassis began to become the space where the subsystems weren't.

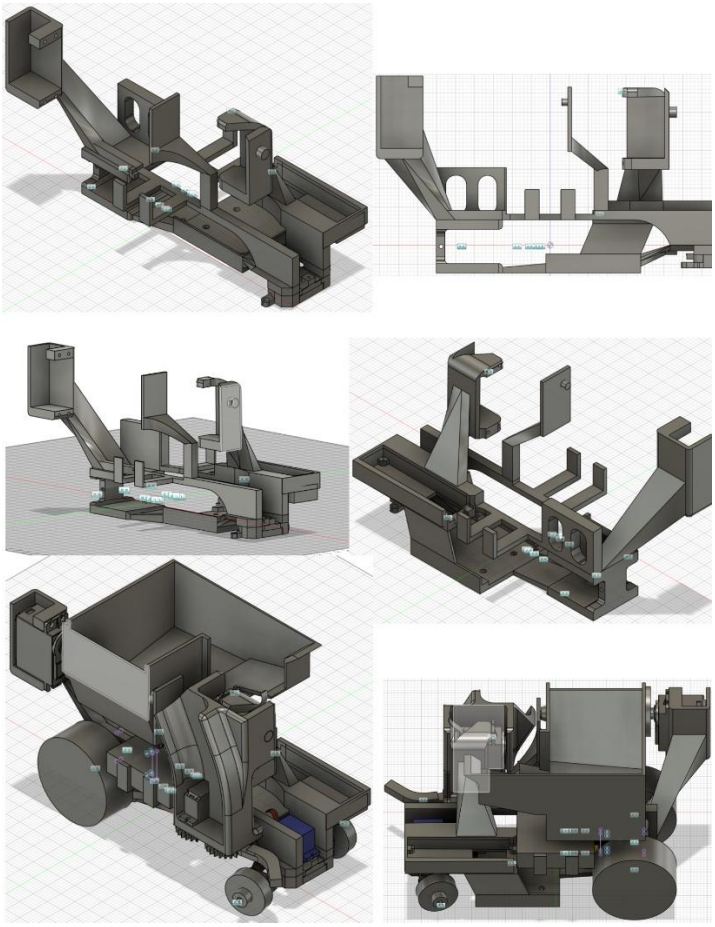
Further development into the electronics system required a capacitor power board to maintain voltage. This system combined with the now required wiring clearance took up a rather large amount of space that further increased the constrains of the design. As the subsystem designs began to come into existence, the volume that these subsystems would need to take up began to fill a significant percentage of the total build volume. As a result, a design overhaul was needed again.

For this next and final phase, all of the components were put into place and the chassis filled the void. The robotic arm and the drivetrain were the big ticket items, without these there is no robot. Then the electronics were put into place where the dumping system would make no sense to be at. This worked very well but had multiple challenges.

The first challenge was drivetrain clearance, the large rear wheels and front steering radius took up a lot of space. To overcome this challenge the drivetrain had to be lengthened. This resulted in the vehicle being too long and sacrificed the crucial turning radius required for the challenge. The second challenge was the battery packs; these components are relatively large and bulky. It was a challenge of its own to both find space for these packs as well as place them in locations that were easy to access for changing the batteries. Luckily, at this point the chassis already had to be lengthened allowing for good placement behind the robotic arm.

After around a dozen hours iterating the component placement and chassis design into such a way that full range of motion for the arm and drivetrain were achieved, the chassis was ready for the dumper. There was only one place for the dumper, a location that allowed the robotic arm to dump the trash into it and fit into the remaining systems. Finally, the dumper was integrated into the chassis and the design was complete.

Figure 6. Final Chassis Design Gallery (Jump, 2026; Alvarado, 2026).



Drivetrain & Locomotion

For the drivetrain, the design had little variation over time. The initial design sketch for the drivetrain was to use 4 Mecanum wheels to allow for maximum maneuverability in tight spaces. However, challenges in programming and the fact that Mecanum wheels are best on flat smooth terrain led to this decision being turned down. The next best option was simple rear wheel drive steering with front wheel steering. Inspired by small RC car simple servo steering, the drivetrain used a servo to control the steering angle through control arms.

Figure 7. *Steering Assembly: Front View (Jump, 2026).*

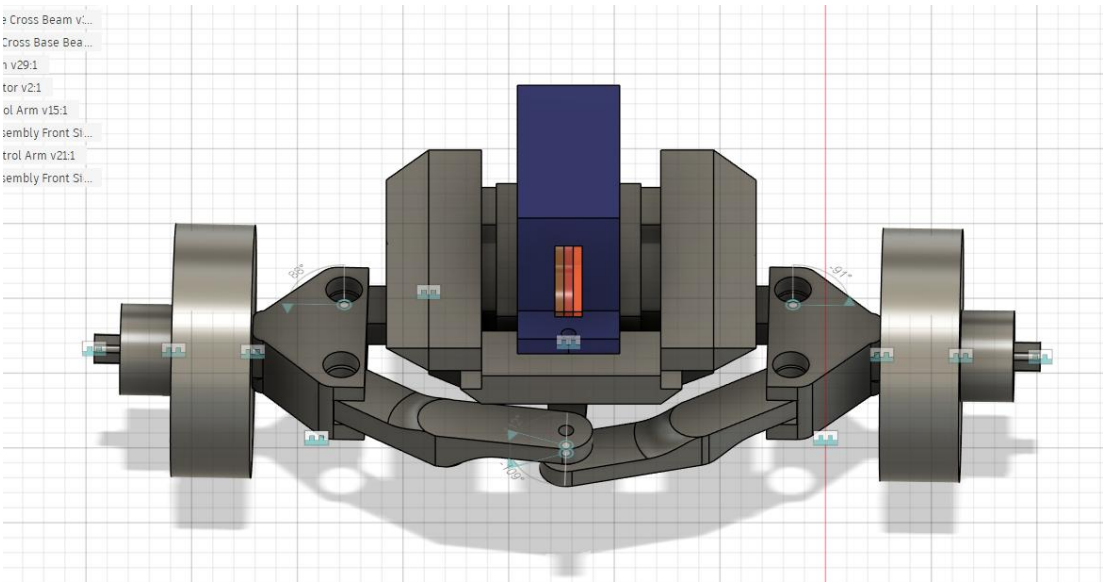
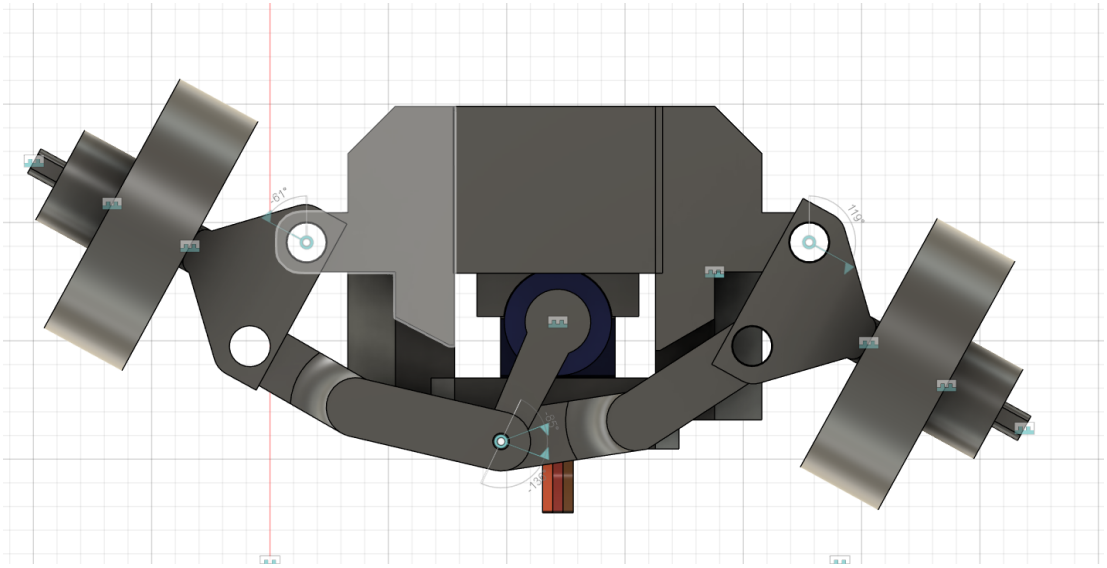


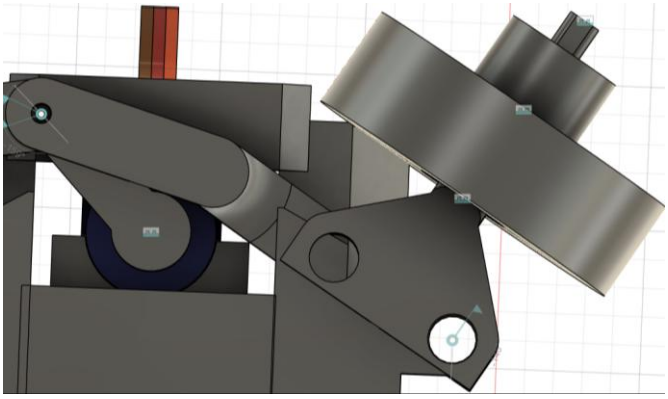
Figure 8. *Steering Assembly: Top-Down View (Jump, 2026).*



This design has pros and cons. The biggest pro is that it is very simple, allowing for more time to be spent working on other subsystems. Mechanically, a servo actuates left and right and the wheels turn with it. As long as the dimensions are right, the servo angle is the steering angle. However, there are some tradeoffs to this design choice. The first is that, the steering angle is limited by the torque of the servo. The force required to turn the wheels is a function of the angle of the servo. At 0 degrees of turning, the servo has the greatest advantage. With this current design, the maximum possible angle under geometric constraints is 53 degrees. Any further and the control arms are parallel to the servo arm, resulting in the angle between the control arms and axle arm also being zero degrees. As a result, no force can be transferred from the servo and the steering locks up. As the steering angle begins to approach 53 degrees, the advantage of the servo approaches zero.

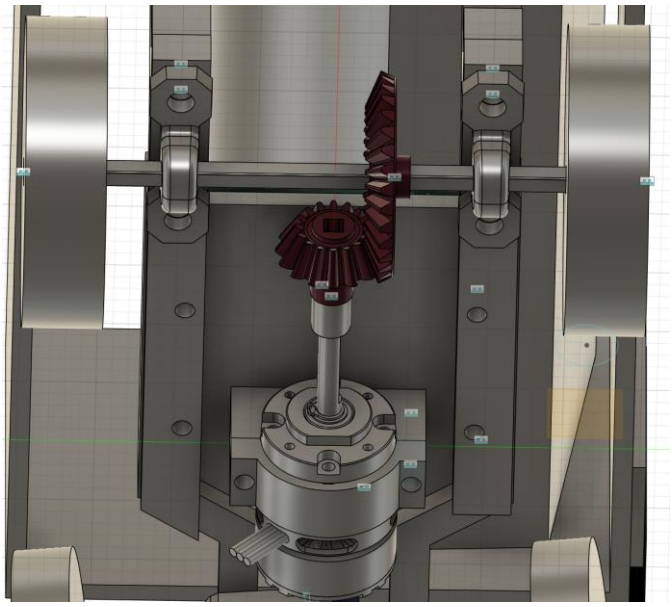
The second tradeoff is the clearance of the steering assembly. Using thick 3D printed plastic doesn't allow much space for maximized steering angles. But a 32-degree max steering angle was achieved but ultimately was not enough for the competition. Every turn was 90 or 180 degrees resulting in a lot of pivoting to navigate these corners. Given the team size and time constraints, this steering system design met the quality required to perform well in the competition.

Figure 9. Steering Radius Closeup (Jump, 2026).



For the rear wheels, there was two iterations of the design. At first, a robust electric motor with a planetary gearbox was going to be used to power the rear wheels with a driveshaft and bevel gear for the wheel axles. Ultimately, the motor drew way too much current from the power supply and was unable to be used. The motor was too robust.

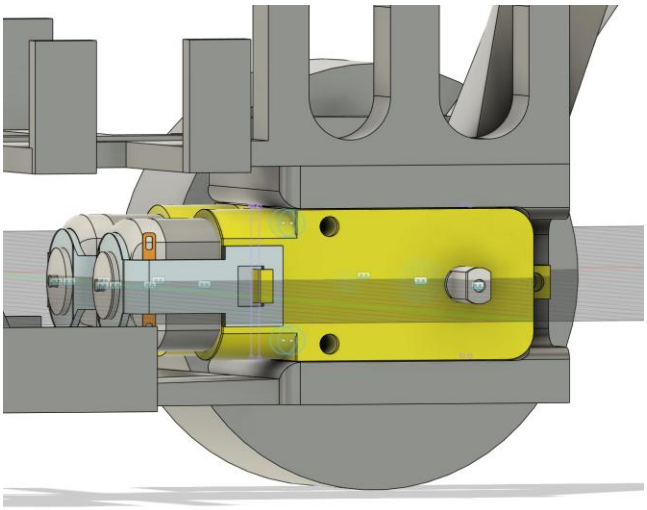
Figure 10. Bottom View, Brushless Motor (Jump, 2026).



The next iteration on the design used two yellow DC motors. These motors took very little power and provided the right amount of torque to move the robot around as needed. The

current draw did not cause any issues with the internal resistance of the power supply resulting in smooth operation. These motors came with their own wheels, and were simple to mount and wire up.

Figure 11. Yellow Motors: Sideview (Jump, 2026).



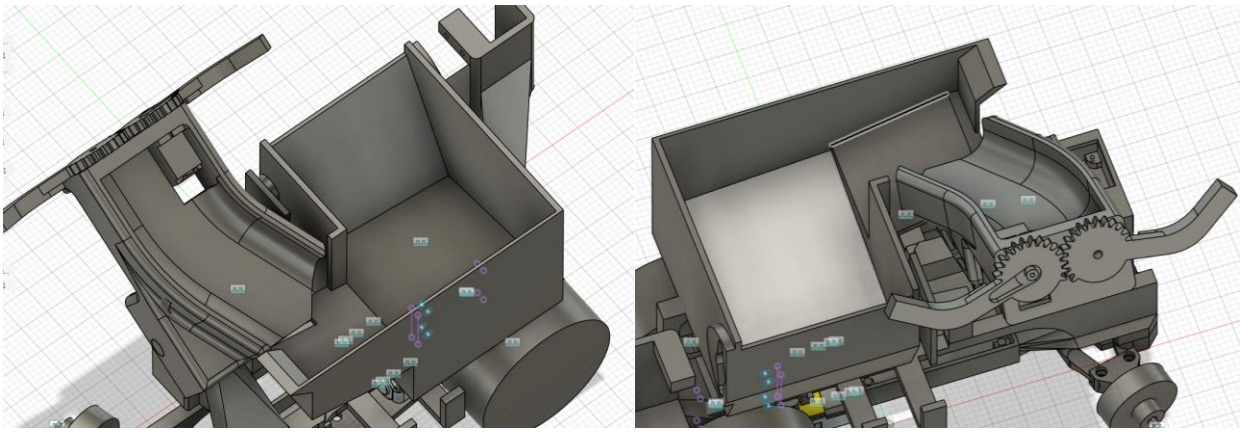
The final design used the yellow DC motors with rubber wheels for the rear drive system. And used completely 3D printed servo steering with rubber glued onto the wheels for the front drive system.

Rotary Dumping Mechanism

The final subsystem to be designed was the dumping system. It needed to perform two functions. The first was that it needed to capture items from the small receptacles picked up by the robotic arm. The second was that it needed to dump these items into the large receptacles. To fulfill these tasks, space was needed for the dumping system. Luckily, the large receptacle required a dumping pivot to be high up. This allowed for much more space due to electrical and drivetrain clearance below.

For the first function, a clever design was used between the arm and the dumper. This design functioned as a slide that guided the items from the small receptacle into the dumper. The process for this was designed very quickly and required good teamwork between the robotic arm designer and the dumping system designer. But through excellent communication and teamwork the task was done effectively.

Figure 12. Assembled Dumping Bucket Top View (Jump, 2026; Alvarado, 2026).



As shown, the contents slide down the arm, into the dumper channel, and into the dumper. This design proved to be very effective. With an arm angle of 35 degrees, nearly any material will slide out of the small receptacle.

The next function of the dumper was to dump the items into a large receptacle. This was done rather easy as the space and height required for dumping was easily achieved. The size of the dumper is simply the remaining space in the build volume. This was the design very fast and satisfying as the constraints were independent of the dumper. The dumper was supported by a servo on the back side and pivoted on a piece of plastic on the other. This design tipped the dumper with an analog control, allowing whatever is inside to slide out smoothly.

Figure 13. Dumping Bucket Pivot Showcase (Jump, 2026).

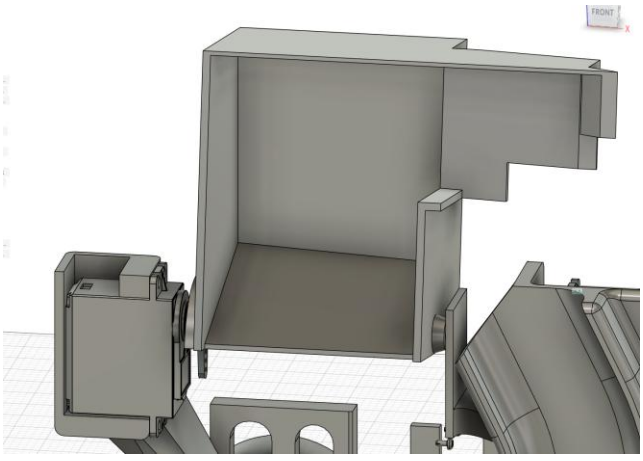
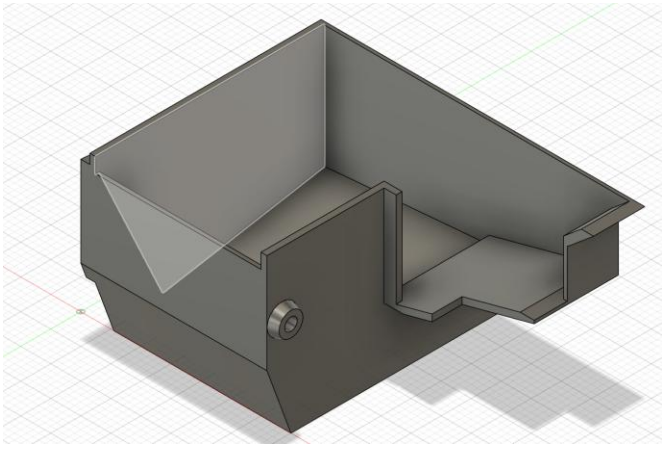


Figure 14. Dumping Bucket Disassembled (Jump, 2026).



Design Tradeoffs Discussion

Several key design tradeoffs were made to balance performance, reliability, safety, and simplicity under the constraints of the ASME SDC environment.

Electrical Systems

The choice of a 6-cell NiMH battery pack was driven primarily by competition requirements and safety considerations. While Lithium Polymer (LiPo) batteries offer higher energy density and discharge performance, they were avoided due to classification concerns and handling complexity. NiMH batteries were selected because they are widely available, robust to handling, and align with the “dry-cell” requirement while providing sufficient current capability for drivetrain and actuator loads. The modular AA form factor also simplified packaging and replacement during competition conditions.

An Arduino-based control architecture was selected for its simplicity, low power consumption, and adequate processing capability for the required teleoperation tasks. Since the system did not require intensive onboard computation or autonomy, higher-performance microcontrollers were unnecessary. The Arduino platform also reduced development complexity and improved debugging speed, which was critical under iterative build and test constraints.

The custom PCB was implemented to reduce wiring complexity and improve mechanical and electrical reliability. By consolidating signal routing, control interfaces, and power regulation into a single board, the system minimized loose connections and improved resistance to vibration and handling during competition. This modular approach also simplified subsystem replacement and troubleshooting under time pressure.

The power architecture intentionally routes battery voltage directly to high-current loads such as the motor driver and high-torque servos. This decision was made to avoid current limitations and voltage drops that would occur if these loads were placed downstream of a regulated buck converter. Instead, regulation was reserved for logic-level systems only. This

separation ensures stable control electronics operation while preserving full power delivery capability for actuation systems, particularly during high-load transient events such as acceleration and steering.

Competition Performance Results

The robot successfully completed one Initial Testing Demonstration and three Head-to-Head Elimination rounds during the ASME SDC 2026 competition. Across these runs, the system demonstrated consistent overall functionality, with performance improving as familiarity with the field conditions increased.

During the Initial Testing Demonstration, the team was still adapting to the layout and operational characteristics of the playfield. The NiMH battery system performed reliably throughout extended operation, with the most stable performance observed when batteries were fully charged prior to each run. Early testing revealed intermittent connection issues in several Dupont-style signal connectors, which resulted in minor control interruptions. These were identified and addressed by transitioning critical connections to soldered joints following the initial round.

Mechanical performance was generally stable; however, reduced wheel traction was observed due to loose particulate material (sawdust-like debris) on the field surface, which affected grip under acceleration and turning. The steering system functioned as intended but exhibited a turning radius that was marginal for the tight urban course geometry, limiting maneuverability in constrained intersections.

The placement of the manipulation arm on the right side of the robot introduced operational constraints when accessing certain collection bins, reducing optimal approach angles in specific field configurations. This limited efficiency in some material collection scenarios, particularly in densely packed or awkwardly oriented zones.

Despite these limitations, the robot advanced through the elimination bracket and reached the final head-to-head matchup against the top-seeded team. Final ranking placement was determined based on cumulative scoring across rounds, resulting in a third-place overall finish.

Overall, the system demonstrated strong reliability in power delivery and control responsiveness, with primary performance limitations arising from mechanical traction, steering geometry constraints, and subsystem placement tradeoffs.

Conclusion

The team designed, built, and competed with a remotely controlled waste collection robot for the 2026 ASME Student Design Competition. The final system combined a custom PCB and power distribution board, an Arduino Nano Every running an iBUS-based teleoperation program, a 3D printed PETG chassis, a rear wheel drive and front servo steering drivetrain, and a rotary dumping mechanism that worked together with the robotic arm. The robot

completed one Initial Testing Demonstration and three Head-to-Head Elimination rounds, reached the final matchup against the top-seeded team, and finished third overall.

The electrical system held up well. Separating the high-current actuator power from the regulated logic power kept the control electronics stable, and the NiMH pack with bulk capacitance handled motor and servo loads without resets. The main performance limits were mechanical: a marginal turning radius for the 90 and 180 degree turns on the course, reduced traction on sawdust-like debris, and an arm placement on the right side that made some bins awkward to reach. The Dupont connector failures in the first round were fixed by soldering the critical connections, which is a lesson worth carrying into future builds.

The biggest takeaway was how much the chassis design changed once it was built around the components instead of the components being fit into the chassis. Doing that earlier would have saved several redesigns. Overall, the project met its goals of building a reliable, repairable, and competitive robot within the size, power, and safety limits of the competition.

Future Improvements

- **Steering and turning radius:** Redesign the steering geometry for a larger maximum angle, or switch to a differential drive layout to allow tighter pivoting in intersections.
- **Traction:** Use wheels with a more aggressive tread or a softer compound, and consider a wheel cleaning or debris-clearing feature for dusty playfields.
- **Arm placement:** Mount the arm closer to the centerline of the robot so it can reach bins from more approach angles.
- **Connections:** Use locking or soldered connectors for all critical signal and power lines from the start, instead of Dupont headers.
- **Electrical protection:** Add an inline fuse and reverse polarity protection to the battery input, and verify servo supply voltage against datasheet limits at full battery charge.
- **Failsafe verification:** Test and confirm receiver failsafe behavior on loss of signal, and add a frame timeout in the Arduino code so the motors stop even if the receiver holds its last values.
- **Documentation:** Record a full power budget and measured voltage under load, and keep a complete bill of materials as parts are added.

Appendix

Figure 15. Arduino Nano Every Code Version 10.0.0 (Hibpshman, 2026; SwapnilNimbalkar, 2020).

```ino
#include <Servo.h>
const int NEUTRAL_US = 1500;
const int DEADBAND   = 100;
const int SMOOTH_W   = 8;
// Motor pins
const uint8_t MOTOR_FWD = 9;
const uint8_t MOTOR_REV = 10;
#define IBUS_BUFFSIZE    32
#define IBUS_MAXCHANNELS 10
uint8_t  ibus[IBUS_BUFFSIZE] = {0};
uint16_t rcValue[IBUS_MAXCHANNELS];
uint8_t  ibusIndex = 0;
Servo steering;     // D2  — CH1, MG90S ±35°
Servo aux;          // D4  — CH3
Servo gripperServo; // D3  — CH5, MG90S 0–90°
Servo shoulderServo;// D5  — CH6, MG995 slowed
int lastThrottle = NEUTRAL_US;
int shoulderPos  = 60;
void driveMotor(int us) {
  int centered = us - NEUTRAL_US;
  if (abs(centered) < DEADBAND) {
    analogWrite(MOTOR_FWD, 0);
    analogWrite(MOTOR_REV, 0);
    return;
  }
  if (centered > 0) {
    int pwm = map(centered, DEADBAND, 500, 0, 255);
    analogWrite(MOTOR_FWD, constrain(pwm, 0, 255));
    analogWrite(MOTOR_REV, 0);
  } else {
    int pwm = map(-centered, DEADBAND, 500, 0, 255);
    analogWrite(MOTOR_FWD, 0);
    analogWrite(MOTOR_REV, constrain(pwm, 0, 255));
  }
}
void setup() {
  Serial.begin(115200);
  Serial1.begin(115200);
  pinMode(MOTOR_FWD, OUTPUT);
  pinMode(MOTOR_REV, OUTPUT);
  analogWrite(MOTOR_FWD, 0);
  analogWrite(MOTOR_REV, 0);
  steering.attach(2);
  aux.attach(4);
  gripperServo.attach(3);
  shoulderServo.attach(5);
  shoulderServo.write(shoulderPos);
  Serial.println("Ready. Waiting for RC...");
}
void loop() {
  if (Serial1.available()) {
    uint8_t val = Serial1.read();
    if (ibusIndex == 0 && val != 0x20) return;
    if (ibusIndex == 1 && val != 0x40) { ibusIndex = 0; return; }
    ibus[ibusIndex++] = val;
    if (ibusIndex >= IBUS_BUFFSIZE) {
      ibusIndex = 0;
      // Checksum
      uint16_t checksum = 0xFFFF;
      for (int i = 0; i < 30; i++) checksum -= ibus[i];
      uint16_t rxChecksum = ((uint16_t)ibus[31] << 8) | ibus[30];
      if (checksum != rxChecksum) return;
      // Parse channels
      int high = 3, low = 2;
      for (int i = 0; i < IBUS_MAXCHANNELS; i++) {
        rcValue[i] = ((uint16_t)ibus[high] << 8) | ibus[low];
        high += 2; low += 2;
      }
      // ===== CH1: Steering (MG90S, ±35°) =====
      steering.writeMicroseconds(constrain(rcValue[0], 1080, 1820));
      // ===== CH2: Brushed motor =====
      int raw = constrain(rcValue[1], 1000, 2000);
      int throttle = (raw * SMOOTH_W + lastThrottle) / (SMOOTH_W + 1);
      lastThrottle = throttle;
      driveMotor(throttle);
      // ===== CH3: Aux =====
      aux.writeMicroseconds(constrain(rcValue[2], 1000, 2000));
      // ===== CH5: Gripper (MG90S, 0–90°) =====
      int gripAngle = map(constrain(rcValue[4], 1000, 2000), 1000, 2000, 0, 120);
      gripperServo.write(gripAngle);
      // ===== CH6: Shoulder (MG995, 0–120°, slowed) =====
      int targetPos = map(constrain(rcValue[5], 1000, 2000), 1000, 2000, 0, 120);
      if (shoulderPos < targetPos) shoulderPos++;
      else if (shoulderPos > targetPos) shoulderPos--;
      shoulderServo.write(shoulderPos);
    }
  }
}
```

Table 1. Bill of Materials (Hibpshman, 2026).

Item	**Quantity**	**Unit Cost ($)**	**Total Cost**	**Notes**
Arduino Nano Every	1	18	18	Microcontroller
FS-IA6B/X TX/RX	1	45	45	Communications
Geared DC Motor	2	1	2	Locomotion
AA Battery Cage	1	8	8	Power Source
Servo (SG90)	1	2	2	
Servo (MG995)	2	4.5	9	
PCB	1	5	5	
PDB	1	2	2	
Misc. Wires	-	2	2	
3D Printed Parts	-	4	4	Structure
			$97	

References

Alvarado, D. (2026, April).

Hibpshman, S. (2026, January). Cheney, Washington, United States of America.

Hibpshman, S. (2026, February 26). ANALYSIS OF BATTERY VOLTAGE UNDER LOAD. Cheney, Washington, United States of America.

Jump, Z. (2026, April).

SwapnilNimbalkar. (2020, December 4). *DIY 10 CH (iBus to PWM) Receiver for FlySky I6 and iA6b Receiver* . Retrieved from YouTube: https://www.youtube.com/watch?v=HpChhPt1CCQ&t=196s